Mohamad Abadi

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EDUCATION

University College London (UCL) – London, United Kingdom Sep. 2024 – Sep. 2025

- MSc in Chemical Process Engineering (Distinction)
- Thesis: *Modelling Tyre Pyrolysis in a Rotary Kiln using Induction-Heated Susceptors*
- Modules: Process Engineering Modelling and Design, Advanced Process Optimisation, Energy Systems

American University of Beirut (AUB) – Beirut, Lebanon Sep. 2019 – May. 2024

- BEng in Chemical Engineering with a minor in Chemistry (Graduating GPA: 3.8/4.0)
- Placed on Dean's Honor List in the last four semesters

PROFESSIONAL EXPERIENCE

Operations Manager – BNS Real Estate LTD – London, United Kingdom Aug. 2024 – Present

- Manage end-to-end operations across 30+ tenanted residential properties, coordinating tenants, landlords, contractors, building management, and a small internal team
- Reduced contractor spend by 40% by benchmarking and replacing the incumbent maintenance provider, and cut monthly storage costs by 73% through Excel-based space optimisation
- Managed property projects and incidents end to end, including renovation planning, AutoCAD layouts, insurance claims, contractor coordination and temporary tenant relocation

Junior Consultant – FREE Energy – Remote Sep. 2023 – Jan. 2025

- Delivered energy audits to ISO 50002, EN 16247 and ASHRAE 211 across 20+ public and private sector sites in the UAE, Saudi Arabia and Lebanon, spanning hospitality, government, education, commercial and industrial facilities
- Built consumption baselines from utility bills and distribution board logger data, including deriving cost of energy from generator fuel curves at off-grid sites with no tariff
- Developed and costed energy conservation measures across HVAC, chillers, lighting, pumps and BMS, priced through bills of quantities and appraised on NPV, IRR and payback alongside avoided emissions in tCO_2e, identifying up to 579,000 kWh and AED 116,000 in annual savings at a single site
- Designed and delivered energy efficiency training in English and Arabic, including a train-the-trainer programme for facilities staff and a seven-competency building insulation course with trainer manuals and assessments

Lab Research Intern – University of Barcelona – Barcelona, Spain Jun. 2023 – Aug. 2023

- Conducted 24 systematic experiments involving the preparation, processing, and testing of polymer-based materials
- Used FTIR and viscosity analysis to troubleshoot failed reactions and optimise catalyst concentration, achieving successful CAP synthesis after initial failed trials
- Produced biodegradable cellulose acetate propionate (CAP) films as a sustainable alternative to petroleum-based packaging materials

EXTRACURRICULAR ACTIVITIES & LEADERSHIP EXPERIENCE

Senior Editor & Social Media Representative – AUB Outlook – Beirut, Lebanon Sep. 2020 – May. 2024

- Progressed from writer to Senior Editor, managing a team of 8+ writers and editors across sports and technology coverage
- Authored 10+ published articles, covering university sports, technology trends, and student life
- Launched and filmed a series interviewing sports team captains, increasing university sports engagement and visibility

Class Representative – American Institute of Chemical Engineers – Beirut, Lebanon Sep. 2022 – Sep. 2023

- Planned a 5-day industrial trip for 40 students to visit 3 chemical engineering plants in Jordan
- Communicated classmates' concerns and requests to the faculty chairperson, resulting in curriculum changes
- Directed and starred in 3 promotional videos for AIChE's social media campaign, increasing engagement by 80%

PROJECTS

Operations & Analytics Dashboard Sep. 2026 – Oct. 2026

- Defined business rules and KPI logic for an AI-assisted Flask/SQLite system replacing spreadsheet reporting across bookings, expenses, and documents, with one active revenue source per property-month to prevent double counting

Master's Thesis – Modelling Tyre Waste Pyrolysis in a Rotary Kiln Nov. 2024 – Sep. 2025

- Developed two 1-D MATLAB kiln models comparing conventional and induction-assisted heating, reducing modelled specific energy demand from 1,400 to 1,270 kJ/kg while identifying a trade-off as conversion decreased from 100% to 88.5%

Renewable Energy Platform and Anaerobic Digestion Plant in Lebanon Sep. 2023 – May. 2024

- Co-designed an integrated solar, wind, wave and biogas system in a team of five, sizing an anaerobic digestion plant for 2,000 tonnes of organic waste a day across 27 digesters and optimising 27.6 MW of generation, 241.8 GWh a year

SKILLS & INTERESTS

- **Technical Skills:** Excel, Microsoft Office, Python, Aspen HYSYS & Plus, MySQL, MATLAB, GAMS, gPROMS, AutoCAD, HelioScope
- **Languages**: Arabic (Native), English (Fluent)